
System1Bench: Typed Decisions Across Domains and Matched Interventions

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Abstract

Typed decision interfaces make finite choices, binary probabilities and ordinal judgments that applications can consume directly. We introduce System1Bench, a cross-domain evaluation of five model–adapter systems over 15 source collections, 28 main suites and eight matched controls. A source atlas distinguishes application domains, decision functions and reference provenance; heterogeneous labels are never pooled into one intelligence score. Paired measurements separate prediction discordance from correctness turnover: on Banking77, candidate reversal changes 38% of Laya Multilingual decisions despite zero net accuracy change. In a frozen 288-state executable-policy experiment, orthogonal display and answer-code interventions expose cancellations invisible to ordinary reversal. A further held-out 96-state wording test finds that making a conjunction explicit changes Jev’s English XOR-case accuracy from 7/48 to 13/48, but its Chinese accuracy from 46/48 to 13/48. The latter result rules out a language-independent wording explanation of the earlier routing pattern. A separate 98,304-decision local experiment measures resident-GPU performance; hosted latency remains a distinct boundary. Raw predictions, executable references, paired uncertainty estimates and figure code support an auditable evaluation of correctness, stability and efficiency at a fixed downstream decision interface, rather than of whole-system task completion.

1 Introduction

A decision interface can appear reliable under aggregate accuracy while changing its answers when the same candidate set is presented in a different order. This discrepancy matters when an application routes a request, triggers a review, or selects an action from a finite schema. The application consumes the selected action, rather than the model’s average score over unrelated examples. Evaluation must therefore ask both whether a decision matches its reference and whether equivalent presentations preserve that decision.

System1Bench studies interfaces that map a supplied state and typed questions to structured answers. The motivating systems include Laya and Jev, alongside language models constrained to the same answer spaces [Convai Innovations, 2026, TypeSafe AI, 2026]. The term “System 1” identifies this product/interface category. Our unit of comparison is a *model plus adapter on a fixed downstream decision*, not an entire deployed workflow. Whole-system success also depends on retrieval, tools, human review and action execution, none of which we measure.

The motivating empirical result is concrete. On the same 100 Banking77 control states, reversing candidate presentation changes 48% of Laya English predictions while increasing reference accuracy by nine percentage points. Laya Multilingual changes 38% of predictions with zero net accuracy change. A leaderboard displaying only the two marginal accuracies would omit most of the behavioral change. The comparison is not a universal degradation claim: it identifies sensitivity that has no fixed direction in aggregate correctness.

Our approach combines a cross-domain source atlas with matched perturbation diagnostics. Sources differ in application domain, decision function, reference provenance, output type, candidate car-

dinality, language, context length and the number of questions sharing a state. We preserve these distinctions instead of averaging unlike references. Paired tests compare the same semantic unit across encodings, and same-order repeats distinguish changed presentation from ordinary execution variability. Separate operational experiments measure resident-model completion and hosted calls under their respective observable boundaries.

This paper makes three contributions. First, it maps 15 sources to non-exclusive application domains and decision functions while retaining reference provenance and sample identity. Second, it separates net correctness, correctness turnover and raw prediction discordance under paired interventions, including a frozen policy diagnostic and an orthogonal answer-code control. A subsequent held-out wording experiment tests and qualifies a mechanism suggested by exploratory inspection. Third, it supplies a reproducible quality-and-efficiency record with frozen local timing controls and a separately pinned hosted API track. The historical analyses are exploratory: their outputs existed before the diagnostic questions were finalized; the two generated follow-ups were frozen before their respective inference runs.

The central implication is methodological. Stable accuracy, stable decisions and efficient execution are distinct properties. A practical benchmark should reveal which of them a system satisfies on a particular workload.

2 Related work

Behavioral and multidimensional evaluation. Behavioral testing organizes model evaluation around capabilities and controlled transformations rather than a single held-out score. CheckList formalizes minimum-functionality, invariance and directional-expectation tests for NLP systems [Ribeiro et al., 2020]. System1Bench adopts this separation at a finite-decision interface: candidate order, language realization and irrelevant state fields define invariance-style comparisons, while an operative fact change defines a directional test with recomputed executable labels. The relevant unit is a shared state cluster, not every transformed row treated as independent evidence.

HELM broadens evaluation across scenarios and metrics, emphasizing that accuracy alone leaves substantial behavior unmeasured [Liang et al., 2022]. Our setting is narrower: finite typed decisions under explicit candidate and rubric contracts. This narrower interface enables exact prediction matching and shared-state perturbations across native decision heads and code-scored language models. It also requires preserving distinctions between dataset annotations, authored policy references, synthetic teachers and programmatic labels. A large count of decisions cannot substitute for an explicit account of what those references establish.

Recent domain-specific studies reinforce this need. Legal-document evaluation of Jev and language models reports that equal aggregate scores can mask different persistent decisions, corrections and regressions [Zhang et al., 2026]. Computational-social-science text-annotation work evaluates many tasks and models with quality, calibration, routing and cost criteria [Ibrahim and Zaki, 2026]. We do not claim that instability or multidimensional evaluation is new. Our complementary contribution is a common *typed decision interface* across deliberately different domains, with paired representation controls, explicit correctness turnover, reference provenance and separate operational boundaries. Cross-domain breadth is useful only when these distinctions survive aggregation.

Finite-choice presentation. Prior work directly documents that language models are not robust multiple-choice selectors [Zheng et al., 2023]. Candidate-order sensitivity is therefore an established problem, not a novel phenomenon claimed here. System1Bench extends the measurement question to native typed heads, shared-state correctness/discordance accounting, and operational interfaces. Its fresh LLM codebook control separates display order from the assignment of semantic labels to answer codes, whereas ordinary reversal changes both. This distinction prevents an adapter-level effect from being attributed to architecture.

Cross-lingual and contextual behavior. XNLI supplies cross-lingual inference examples with shared labels, supporting paired evaluation of language realizations [Conneau et al., 2018]. We use the source alignment to compare English and Chinese under the existing question wrapper. Translation realization and wrapper language are part of this intervention; the paired result describes the implemented interface. CLINC150 similarly separates in-scope intent classification from explicit

out-of-scope prediction [Larson et al., 2019]. That distinction is operationally relevant because a finite choice among known actions does not by itself supply a reliable reject option.

Long-context results depend on the position of the relevant information as well as the nominal context budget. Lost in the Middle demonstrates this dependence in long-context language-model tasks [Liu et al., 2024]. Our needle diagnostic retains source length and insertion position, but also separates the finite-choice intent question from a binary damage question attached to the same state. Grouping by the original needle preserves dependence among length and position variants. The diagnostic concerns explicit fact retrieval, rather than complete long-horizon decision competence.

Selective prediction. Selective classification evaluates the trade-off between making predictions and rejecting uncertain cases [Geifman and El-Yaniv, 2017]. A useful finite-decision model should ideally concentrate its errors among low-confidence cases. We plot empirical risk against coverage within each source, accepting tied confidence values together. The common ranking statistic is maximum class probability. Native decision confidence and a language model’s softmax over candidate codes need not share calibration semantics, so selective curves characterize observed error ranking under each adapter rather than interchangeable confidence thresholds.

Inference measurement. MLPerf Inference distinguishes operating scenarios and requires performance to be considered with quality constraints [Reddi et al., 2020]. PagedAttention evaluates serving efficiency under an explicitly specified workload and system configuration [Kwon et al., 2023]; Sarathi-Serve studies the throughput–latency trade-off in LLM inference [Agrawal et al., 2024]. We use these principles to declare timing boundaries, workload shapes, hardware, warm-up and replication. Our local experiment measures an uncached state-to-structured-answer call in a resident process. It excludes model loading, network transport and queueing. Hosted Jev calls expose a different boundary and remain a separate track. This separation is necessary for meaningful interpretation of model-adapter timing rather than a mixed comparison of service and local execution.

3 Task taxonomy and measurement design

3.1 Typed contract and reference provenance

An input comprises a state s_i and a question map Q_i . Each question specifies an instruction, an output primitive and, when applicable, a candidate set or ordinal rubric. A *choice* selects a finite label, a *noul* returns a binary probability, and a *score* returns a distribution over ordered levels together with its expectation. Model-facing payloads contain only state and questions; reference labels and hidden source metadata are excluded by an explicit whitelist.

The historical collection contains 15 source datasets or collections, instantiated as 28 main suites and eight repeat/reversal controls. Figure 1 maps each source to an application domain and non-exclusive decision functions: semantic classification, routing, evidence use, ordinal judgment, safety, workflow, explicit rejection and retrieval. These functions are descriptive tags, not equal-weight components of a latent intelligence score. A suite additionally records primitive, language, candidate count, state/question multiplicity and reference type. Main-suite variants can share underlying examples: BoolQ and SST5, for example, each have two output encodings.

SOURCE	DOMAIN	DECISION FUNCTION								REF. SUITES	
		Semantic	Routing	Evidence	Ordinal	Safety	Workflow	Reject	Retrieve		
LANGUAGE AND AFFECT											
AG News	News	●	•	•	•	•	•	•	•	D	1
Emotion	Social text	●	•	•	•	•	•	•	•	D	1
SST-5	Reviews	●	•	•	●	•	•	•	•	D	2
SERVICE AND ROUTING											
Banking77	Banking support	•	●	•	•	•	•	•	•	D	1
MASSIVE	Assistant commands	•	●	•	•	•	•	•	•	D	2
CLINC150/OOS	Assistant intents	•	●	•	•	•	•	●	•	D	1
EVIDENCE AND INFERENCE											
BoolQ	General QA	•	•	●	•	•	•	•	•	D	2
XNLI	Natural inference	●	•	●	•	•	•	•	•	D	2
TurtleBench	Puzzle evidence	•	•	●	•	•	•	•	•	D	1
SAFETY											
Prompt injections	Prompt security	•	•	•	•	●	•	•	•	D	1
Aegis2 prompts	Content safety	•	•	•	•	●	•	•	•	H	1
TYPED WORKFLOW FIXTURES											
Typed decisions	Mixed enterprise	•	•	•	●	●	●	•	•	T	1
JevBench	Mixed fixtures	•	•	●	●	•	●	•	•	A	3
ReflexBench	Product workflow	•	•	•	•	•	●	•	•	A	1
Jev-Laya	Mixed workflow	•	●	●	●	●	●	•	●	T/P	8

● Task appears in source D dataset · H human prompt · T synthetic teacher · A authored/reviewed · P programmatic

Figure 1: Cross-domain source atlas. Rows are 15 source collections, grouped by application domain; filled cells mark non-exclusive decision functions. S is the number of evaluated main suites, not independent datasets. Reference codes distinguish dataset labels (D), human prompt annotations (H), synthetic teachers (T), authored or AI-reviewed cases (A) and executable labels (P). The atlas classifies workload coverage, not model quality.

References are classified as dataset-provided, human prompt annotations, authored or AI-reviewed policy cases, synthetic-teacher outputs, or programmatic labels. Exact agreement with a synthetic teacher is reported as teacher agreement. The source manifest pins downloaded artifacts and preparation code; request fingerprints preserve dictionary insertion order because candidate order is an experimental variable. Original model outputs and input audits remain immutable.

3.2 Correctness, discordance and paired effects

Let $\hat{y}_i^{(a)}$ be the canonicalized finite prediction under condition a , and let y_i be the reference. Invalid responses count as incorrect. Define $c_i^{(a)} = \mathbf{1}[\hat{y}_i^{(a)} = y_i]$. For a semantics-preserving intervention $a \rightarrow b$, we report

$$\Delta_{a \rightarrow b} = \frac{1}{n} \sum_i (c_i^{(b)} - c_i^{(a)}), \quad D_{a \rightarrow b} = \frac{1}{n} \sum_i \mathbf{1}[\hat{y}_i^{(a)} \neq \hat{y}_i^{(b)}]. \quad (1)$$

These quantities answer different questions. If N_{01} counts incorrect-to-correct changes and N_{10} correct-to-incorrect changes, then $\Delta = (N_{01} - N_{10})/n$. The *correctness turnover* $T = (N_{01} + N_{10})/n = |\Delta| + 2 \min(N_{01}, N_{10})/n$ exposes the opposing changes hidden by net accuracy. Discordance D also includes changed predictions that are both incorrect. We retain both transition counts, pair identifiers and source-cluster membership in the analysis artifact.

For candidate reversal, an original condition o , exact-repeat condition r and reversed condition v give the additional contrast $D_{r \rightarrow v} - D_{o \rightarrow r}$. This excess-discordance estimate uses the same clusters in

both terms. It describes the implemented candidate presentation and answer-code mapping jointly. In the LLM adapter, changing candidate order also changes which answer code represents an option.

Uncertainty is estimated with 10,000 paired cluster-bootstrap draws and seed 20260929. The cluster is the source/state unit, including all questions and matched variants. Needle analyses resample the 25 original content units; their 450 derived states are not 450 independent samples. Intervals for historical contrasts are pointwise and descriptive. Complete contrast families, rather than only favorable directions, are retained.

3.3 Compared systems

The local systems are Laya English and Multilingual from the same pinned repository revision, Llama-3.1-8B-Instruct and Qwen3-8B. Laya uses native typed heads. The LLM adapter serializes each question and scores distinct single-token candidate codes with one forward pass, without generated reasoning. Its probabilities are conditional on those candidate logits. Supplied binary criteria and ordinal levels are preserved. All historical local decisions passed complete-input checks; checkpoint file hashes identify the actual local weights.

Hosted Jev. The pinned `jev-1.13.0` endpoint completed 22,934 request executions covering the same 26,450 decisions. 8 decisions failed the declared transport/response validity rule and remain in the accuracy denominator. The API reported 23,841,873 input tokens, corresponding to USD 1.001 at the retrieved public price; unavailable usage on failed calls is not included. Every returned version and answer is retained. Server tokenization and weights are not observable. Hosted client times include the proxy, network and service and are reported separately from resident-GPU measurements.

3.4 Fresh executable-policy diagnostic

The new diagnostic generates 288 unique base states across refund, access-control and routing policies using three fixed data seeds. Original action labels are stratified, and boolean/ordinal marginal distributions are reported rather than assumed balanced. Each policy includes a priority-ordered action, an escalation predicate and a severity rule. The correct outputs are computed directly from visible facts and policy parameters and checked by a separately expressed rules implementation.

Eight matched variants test exact repetition, action-order reversal, an authored Chinese rendering, irrelevant archived records, question paraphrases, equivalent choice encodings, and a one-field counterfactual, alongside the original. The counterfactual must change the correct action; all other variants preserve the reference labels. Generation rejects base states without such a one-field intervention. Thus the diagnostic is a curated compositional-rule distribution. The generator, data fingerprints, scoring contract and intervention scope are frozen before inference. Three generator seeds replicate data generation, not model training.

All five systems complete 6,912 decisions each in this diagnostic. The six predeclared contrasts per model/policy family are excess reversal discordance, Chinese/distractor/paraphrase action correctness effects, and choice-encoding effects on review and severity correctness. We use a max-standardized-deviation bootstrap with 10,000 draws stratified by the three data seeds, providing simultaneous intervals within each six-effect family, not across models or policies. Zero empirical variance is flagged rather than interpreted as population invariance. A five-percentage-point practical discussion threshold was fixed before inference; it is not a universal acceptance criterion.

Each LLM additionally answers 1,440 codebook-control decisions: baseline, exact repeat, display-only reversal, code-only reversal and both. Prompt tests verify that baseline matches the ordinary adapter and both matches its reversed-candidate rendering. The native decision interfaces do not expose an equivalent answer-code assignment. These additional contrasts are descriptive and use pointwise intervals.

3.5 Held-out wording challenge

A post-hoc inspection of the fresh routing results motivated a second, separately frozen experiment. We generated 96 new base states from seeds 404, 505 and 606, disjoint from the 288-state policy set, balancing the four outage/payment truth cells within each seed. Each state has English and Chinese *compact* and *explicit* policy renderings. Within a language, only the first policy clause changes; facts,

Source	Model	Repeat flips	Reversal flips	Δ accuracy
banking77	Laya EN	0.0	48.0	+9.0
banking77	Laya Multi	0.0	38.0	+0.0
banking77	Llama	4.0	65.0	-25.0
banking77	Qwen	1.0	39.0	-18.0
banking77	Jev	1.0	3.0	+1.0
massive_en	Laya EN	0.0	50.0	-2.0
massive_en	Laya Multi	0.0	46.0	-7.0
massive_en	Llama	5.0	51.0	-2.0
massive_en	Qwen	0.0	32.0	-8.0
massive_en	Jev	2.0	10.0	-1.0

Table 1: Matched candidate controls. Flip rates and accuracy differences are in percentage points. Reversal is compared with the same-order repeat; the repeat flip rate uses the corresponding original predictions. Source sizes differ and are shown in Figure 2.

parameters, questions, answer options and executable references are held fixed. The preregistered primary contrast is explicit-minus-compact action accuracy on the 48 exclusive-or states in English; Chinese and the remaining truth cells are reported as secondary. All five systems completed 384 requests each. Intervals use 10,000 paired bootstrap draws stratified by seed and truth cell. This is a controlled wording test on authored policy text, not a human-validated translation study.

4 Representation sensitivity beyond aggregate accuracy

4.1 Candidate order exposes hidden instability

Figure 2 juxtaposes prediction flips and the opposing correctness transitions hidden by their net accuracy change. Across the four historical local systems and four control tasks, reversal flips range from 8.3% to 65.0%; same-order repeats range from zero to 5.0%. These measurements support a presentation-sensitivity finding, rather than a uniform accuracy penalty.

Laya English on Banking77 illustrates the distinction: a 48% flip rate accompanies a nine-point accuracy increase. Laya Multilingual flips 38% of predictions with unchanged accuracy. Llama’s 65% reversal flips exceed its four-percent repeat flips, while its accuracy falls by 25 points. A mean accuracy change neither determines the number of affected decisions nor separates presentation effects from execution variation. Recording the paired transition structure is therefore an evaluation requirement whenever stability matters to the application.

4.2 Output encodings and language realizations

The paired output-encoding effects depend on the task and system. On SST5, choice minus ordinal-score accuracy is +15.0, +6.1, +10.2 and -1.8 percentage points for Laya English, Laya Multilingual, Llama and Qwen respectively. BoolQ effects are smaller and differ in direction. These are comparisons of complete interface encodings: instructions, candidate descriptions and adapter execution can all differ. They identify interface sensitivity, not the causal effect of a particular neural head.

The English/Chinese comparisons share source labels and sample identities. Chinese-minus-English accuracy is negative for every historical local system on both XNLI and MASSIVE under the existing English question wrappers. The magnitude varies substantially by model and source. Figure 6 places paired correctness effects beside prediction disagreement; a smaller average drop can still accompany many changed decisions. The practical implication is to validate the actual multilingual request rendering used by an application rather than transferring an English threshold or score by name.

4.3 Context effects depend on the question

Figure 7 separates the choice and binary questions attached to each needle state. This split changes the interpretation of a pooled retrieval score. Llama retains strong choice retrieval across many lengths and positions while its binary damage question degrades with longer contexts. Laya English exhibits sharply different choice behavior when the relevant content appears at the start rather than

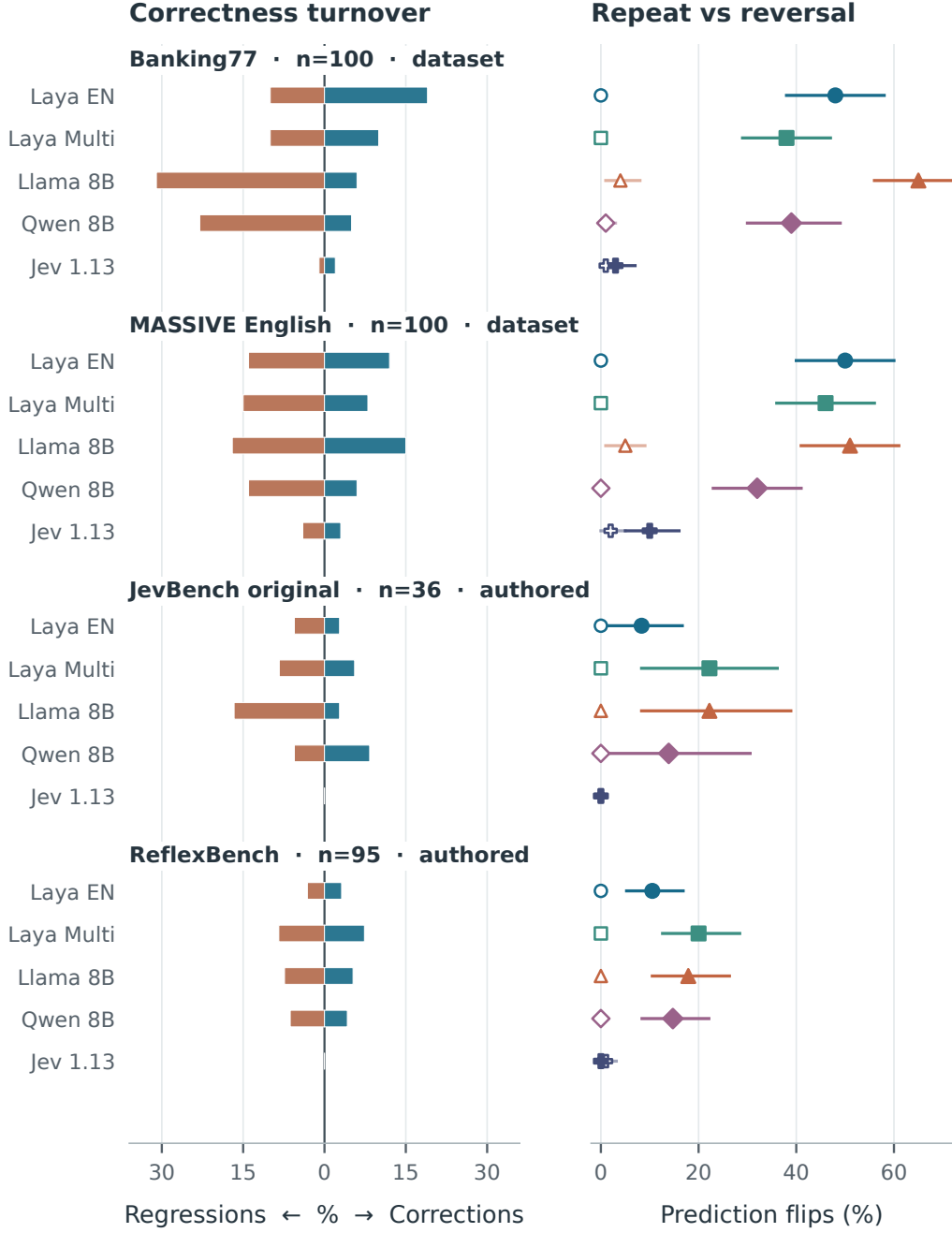


Figure 2: Candidate presentation/code sensitivity. Left: bars show repeat-to-reversal regressions (correct to incorrect) and corrections (incorrect to correct), whose difference is the net accuracy effect. Right: open markers show original-to-repeat flips and filled markers repeat-to-reversal flips with pointwise 95% paired cluster intervals. The control sets contain 100/100, 100/100, 36/18 and 95/95 decisions/source clusters respectively. Banking77 and MASSIVE use dataset references; JevBench and ReflexBench use authored or AI-reviewed references. No averaging across sources is performed.

Policy	Model	Action	Review	Severity	Majority	Joint
Access	Laya EN	34.4	44.8	43.8	34.4	2.1
Access	Laya Multi	29.2	40.6	29.2	34.4	0.0
Access	Llama 8B	34.4	40.6	68.8	34.4	0.0
Access	Qwen 8B	66.7	53.1	79.2	34.4	46.9
Access	Jev 1.13	100.0	89.6	95.8	34.4	100.0
Refund	Laya EN	25.0	77.1	31.2	25.0	0.0
Refund	Laya Multi	25.0	74.0	31.2	25.0	0.0
Refund	Llama 8B	33.3	74.0	59.4	25.0	0.0
Refund	Qwen 8B	25.0	74.0	43.8	25.0	1.0
Refund	Jev 1.13	100.0	100.0	100.0	25.0	100.0
Routing	Laya EN	71.9	69.8	35.4	25.0	28.1
Routing	Laya Multi	24.0	63.5	33.3	25.0	0.0
Routing	Llama 8B	25.0	63.5	59.4	25.0	0.0
Routing	Qwen 8B	49.0	64.6	79.2	25.0	28.1
Routing	Jev 1.13	54.2	100.0	95.8	25.0	29.2

Table 2: Fresh policy reference accuracy (percent). Joint requires both original and one-field-counterfactual actions to be correct. Majority is the original action-label baseline. All rows have 96 paired base clusters; the complete machine-readable analysis includes pointwise cluster intervals and all reference-label distributions.

the middle or end. Qwen’s curves are comparatively stable in this fixed diagnostic. These patterns argue for a length-by-position-by-question analysis rather than a single context-window number or an assumed monotonic degradation law.

4.4 Fresh policy interventions and codebook controls

Table 2 reports the original-state accuracies and joint original/counterfactual action success. The fresh diagnostic covers 96 base states per family, with eight matched variants. Figure 5 retains all six predeclared effects per model and policy, with simultaneous intervals within each six-effect family. The codebook experiment separates display order from answer-code identity for the two LLM adapters.

The fresh results separate invariance from supplied-rule competence. Jev achieves 100% original and joint action success on refund and access policies, but only 54.2% original and 29.2% joint success on routing. Small reversal effects on that policy therefore do not establish correct rule application. Several local model/policy cells are near the action-majority baseline; low intervention effects in such cells can reflect consistently wrong decisions. We report base accuracy and joint counterfactual success alongside every stability diagnostic to expose this ambiguity.

Representation effects also depend on the supplied policy. Jev’s authored Chinese routing rendering increases action accuracy by 45.8 percentage points, with a within-family simultaneous interval of [31.4, 60.2]; the underlying English facts and label IDs remain fixed. This is a difference between two complete policy/question/criteria renderings, not a general Chinese-language advantage. Laya English routing loses 46.9 points under explicitly irrelevant archived records. The full effect family, including null and opposite-direction results, appears in Figure 5.

A post-hoc routing inspection localizes all 44 Jev English action errors to states where exactly one of outage and paying is true: every such state is routed to urgent support, although the policy requires both. All 52 states with matching boolean values are correct. The Chinese rendering is correct on all four boolean combinations. This behavioral pattern motivated the held-out wording test below; it did not isolate the cause because the original language intervention changed multiple text fields together.

The LLM codebook control shows why ordinary candidate reversal can be misleading (Figure 3). For Llama access decisions, reversing display alone changes no predictions, changing only the label-to-code assignment changes 55.2%, and changing both again changes none. On routing, the corresponding values are 0%, 62.5% and 3.1%. These are observed non-additive effects within the fixed adapter: apparent stability under the combined intervention can coexist with sensitivity to code identity. They do not identify an architectural mechanism. The codebook run is a separate inference block, so its repeat and both-change outcomes are not assumed identical to the earlier batch context.

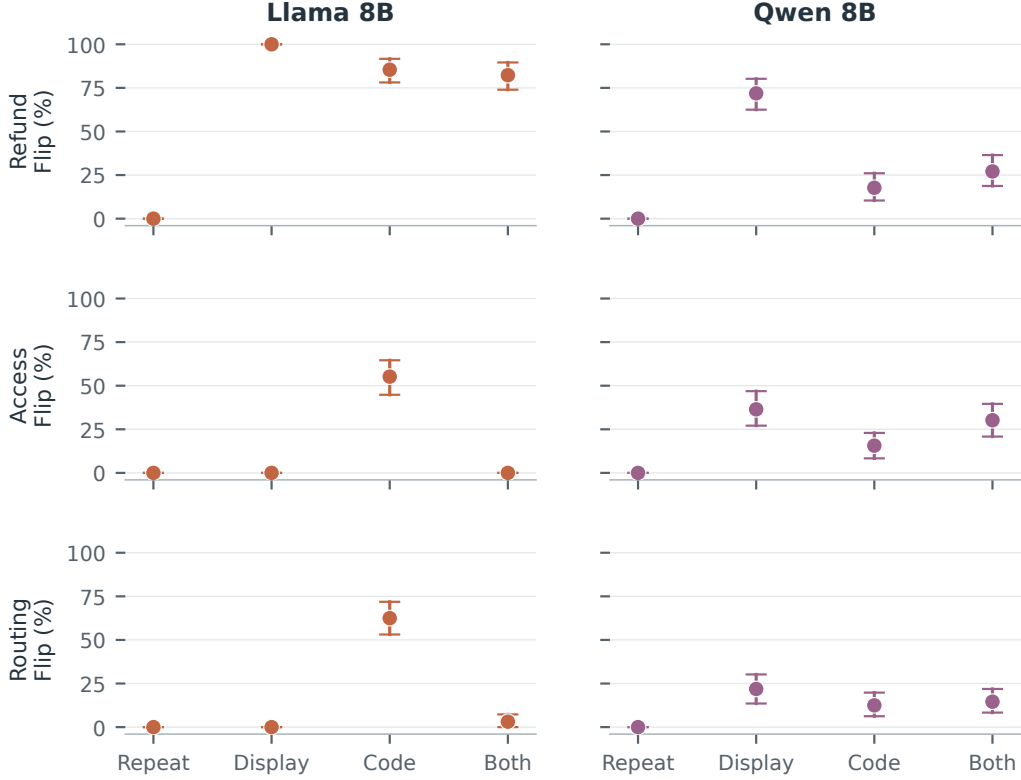


Figure 3: Fresh LLM codebook controls: 96 base action decisions per policy and model, with pointwise 95% paired cluster intervals. Display reverses semantic option order with code identity fixed; Code reverses code identity with semantic order fixed; Both changes both. Ground truth is an executable supplied policy. The code-only sensitivity can be hidden by the combined reversal.

4.5 Held-out wording: a proposed explanation fails to generalize

The new routing cases balance all four outage/payment truth combinations by construction. Figure 4 reports the explicit-minus-compact wording effect on the 48 exclusive-or states, where erroneously interpreting “or” for “and” changes the action. Jev improves from 7/48 to 13/48 in English (+12.5 percentage points; paired 95% CI [2.1, 25.0]), but falls from 46/48 to 13/48 in Chinese (−68.8 points; CI [−81.3, −56.3]). The explicit conjunction is therefore not a language-independent repair. Qwen improves by 29.2 points in English and 52.1 in Chinese, while Laya English worsens by 31.2 and 12.5 points respectively. We report these mixed results because a universal “clearer wording helps” account would be contradicted by the held-out data. The experiment changes one authored policy clause within each language; it does not identify whether the cross-language difference arises from translation quality, tokenization or model training.

5 Operational measurements and confidence ranking

5.1 Local quality and completion time

The local performance experiment evaluates 11 strata, four checkpoints, three batch sizes and eight rounds on the same A100 80GB PCIe, yielding 1,056 cells and 98,304 measured decisions. Each stratum uses 64 fixed measured states and disjoint warm-up states. Model processes are fresh per block; model order is balanced, CPU affinity and thread counts are fixed, and an external observer records GPU, worker-core and SMT-sibling activity.

The timing boundary includes uncached serialization, tokenization, transfer, inference and native structured-answer construction, with GPU synchronization. It excludes model loading, disk output,

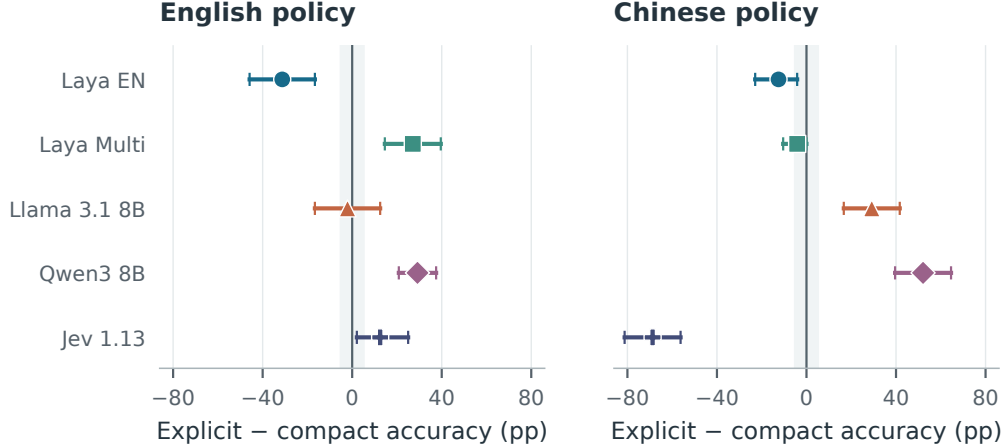


Figure 4: Held-out routing wording challenge on 96 new base states (48 XOR cases). Points are paired explicit-minus-compact action-accuracy effects; whiskers are pointwise 95% bootstrap intervals stratified by generator seed and truth cell. Within each language only the first policy clause changes. English is the predeclared primary contrast; Chinese is secondary. The shaded band marks a ± 5 -point descriptive region, not a significance threshold.

validation, network transport and queueing. Batch-one p50/p95 characterize request completion; batch-eight and batch-32 work divided by prediction time characterize fixed-batch throughput. Round statistics are summarized by their median and an eight-round bootstrap interval. Repeated states increase timing replication, not the effective accuracy sample size.

Figure 8 compares quality and latency within six declared workloads. For BoolQ, median-of-round request p50 values are 10.30, 8.55, 36.86 and 33.51 milliseconds for Laya English, Laya Multilingual, Llama and Qwen respectively. The quality coordinate belongs to the same fixed workload. It is not appropriate to infer a cross-task winner from the horizontal position alone: systems can trade reference agreement for lower completion time, and that trade-off changes with the task.

The host is shared. No foreign GPU compute process or sustained CPU/SMT threshold violation was observed, but six isolated residual-CPU intervals exceeded 15% without meeting the fixed two-interval rejection rule. These intervals remain included. All raw timing records, monitor samples and code/environment hashes are retained, and all 1,056 cells independently reconcile with the generated tables.

5.2 Selective prediction within a source

A software action can be deferred when the model is uncertain. Figure 9 shows empirical accepted-set error as coverage grows under maximum class probability. All tied probabilities enter together, avoiding artificial improvements from an arbitrary within-tie ordering. The curve is computed separately for each model and source; intervals at displayed landmarks condition on the selected threshold.

These curves distinguish two failure modes that accuracy alone conflates: frequent errors at all confidence levels and errors concentrated among uncertain cases. They support a workload-specific investigation of escalation policies. A threshold selected from these test curves would be a post-hoc decision rule, so a deployable policy requires a separate calibration/development split and a held-out check. Candidate-code probabilities are evaluated as observed ranking statistics, not assumed calibrated equivalents of native decision confidence.

6 Limitations and scope

Reference and construct validity. The collection combines heterogeneous source references. Synthetic-teacher agreement, authored policy conformity and programmatic rule correctness are

distinct targets. Static typed decisions do not measure interactive task completion or downstream utility. The fresh policies validate supplied-rule composition; their naturalness and Chinese realization have not received independent human adjudication. AG News and BoolQ belong to known Laya training-task families; these rows cannot establish unseen-task generalization. Other public-source training overlap is unknown.

Intervention and system scope. The local systems are fixed checkpoints with one declared adapter each. Output encodings can jointly change wording and computation, and candidate reversal changes presentation and LLM code assignments together. Findings describe these implementations rather than all decision-model or language-model architectures. Historical diagnostic selection is exploratory; the first generated cases supplied a pre-inference-frozen test of declared transformations. The held-out wording test was frozen after inspecting the first generated results but before its own inference; it tests a motivated hypothesis rather than independently discovering it. Its Chinese clauses lack independent human equivalence validation.

Operational scope. Local measurements use one shared host and fixed-batch inference, not optimized serving stacks or production arrival processes. Approximately one-hertz telemetry cannot exclude brief interference or shared memory-bandwidth effects. Hosted API version IDs and usage are observable, while server weights, tokenization, scheduling and resource allocation are not. The two timing boundaries remain separate.

Statistical scope. Matched variants are dependent and resampled together. Needle analysis has only 25 source clusters. Eight timing rounds support descriptions of these workloads, not rare-tail service guarantees. Pointwise exploratory intervals across many contrasts do not provide simultaneous paper-wide error control. Inference counts are execution totals, not independent-sample counts.

7 Conclusion

System1Bench separates reference agreement, correctness turnover, prediction stability and operational efficiency at a fixed typed decision interface. The cross-domain atlas makes workload coverage and reference provenance explicit without treating unlike datasets as one score. Matched controls show that substantial corrections and regressions can cancel in aggregate accuracy, while the held-out wording test rejects a simple universal explanation of one striking routing result. Question-specific context and workload-specific performance provide further evidence against premature pooling. These are measurements of downstream decision behavior, not whole-system task completion; deployment claims require validation of the complete workflow.

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A Complete source-level results

Source	Laya EN	Laya Multi	Llama	Qwen	Jev
ag_news	94.3	92.4	88.9	86.8	88.6
emotion	59.1	53.0	50.5	54.8	59.0
banking77	55.3	51.2	54.1	66.8	79.9
massive_en	54.1	42.3	57.2	66.3	77.5
massive_zh	30.4	33.2	53.5	62.6	76.0
boolq	84.6	77.7	64.8	83.5	92.6
boolq_choice	83.6	77.4	68.7	83.3	92.2
xnli_en	86.0	81.7	46.2	78.7	85.9
xnli_zh	61.5	74.3	40.5	69.1	73.6
turtlench	42.6	41.6	42.2	44.2	74.0
sst5	34.6	29.5	32.0	45.7	56.5
sst5_choice	49.6	35.6	42.2	43.9	55.9
prompt_injections	70.7	57.8	62.9	63.8	76.7
aegis2_prompt	49.6	57.1	66.8	72.7	82.4
typed_decisions	36.4	34.9	51.2	55.5	73.9
JB: original	70.8	41.7	75.0	83.3	98.6
JB: easy	95.8	89.6	100.0	100.0	100.0
JB: hard	29.7	32.4	34.2	46.8	72.1
ReflexBench	58.9	46.3	60.0	84.2	93.7
JL: triage	62.5	55.7	75.6	80.0	90.0
JL: moderation	67.8	48.4	89.0	77.5	92.7
JL: routing	64.2	51.1	61.3	83.7	88.8
JL: claims	90.0	80.0	61.7	98.7	100.0
JL: reviews	70.7	42.3	83.0	87.0	87.7
JL: guard	60.6	33.2	88.0	83.2	93.8
JL: multilingual	57.8	62.5	94.5	98.4	100.0
clinc150_oos	55.7	64.8	55.2	67.8	89.5
JL: needle	50.2	47.4	77.9	87.8	95.2

Table 3: All 28 main suites, in fixed taxonomy order. Entries are percentage reference agreement; synthetic/authored sources retain their respective reference semantics. These rows are not independent benchmarks and are not averaged. Failure cases remain in denominators. Full per-source cluster intervals, type/language slices and provenance are provided by the machine-readable artifact.

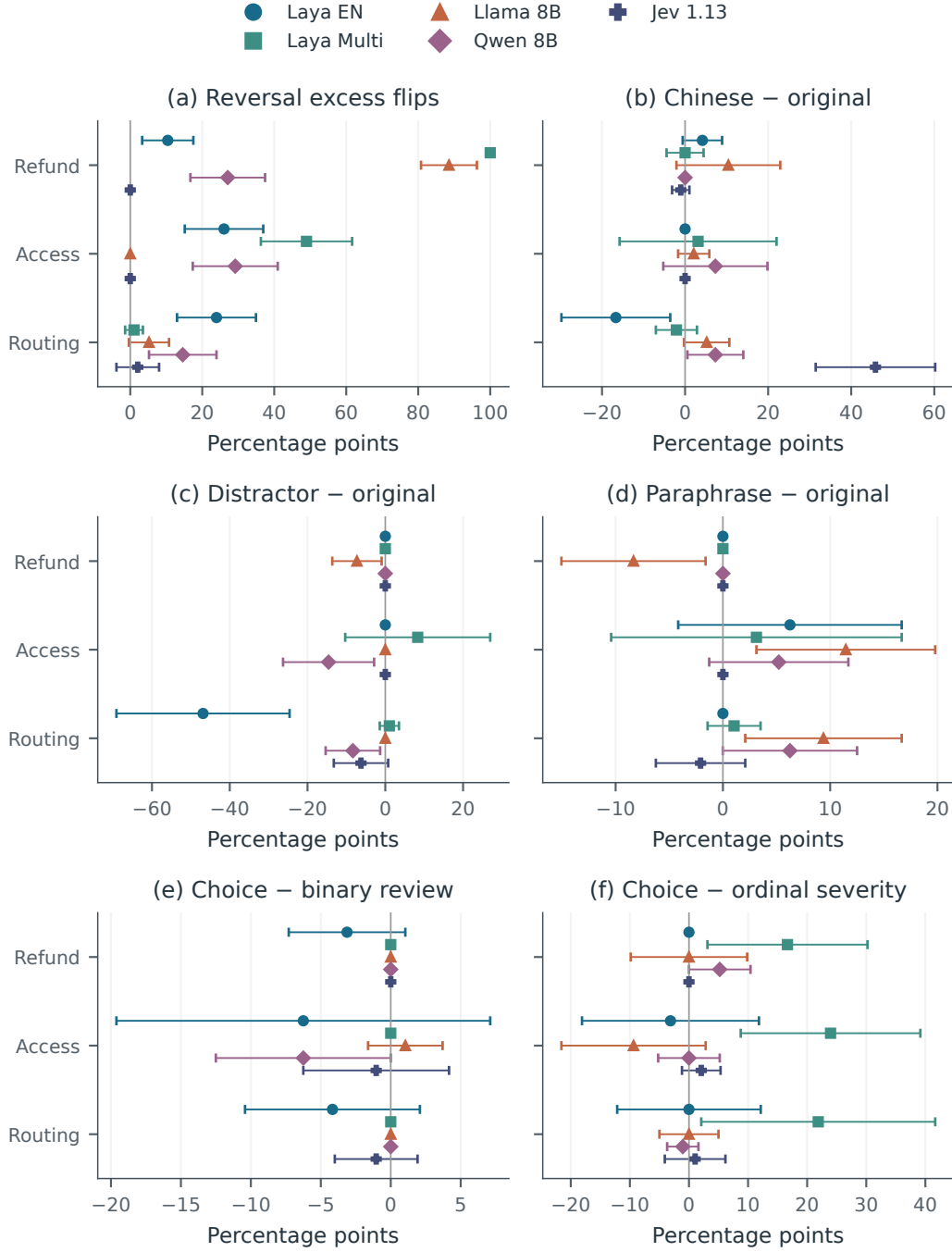


Figure 5: All six predeclared fresh-policy effects, with 96 base clusters per policy. Whiskers are simultaneous 95% intervals within the six-effect family of each model/policy, using 10,000 bootstrap draws stratified by generator seed. They do not provide paper-wide coverage. Panel (a) measures excess action flips; (b)–(f) measure correctness differences for the stated primitive. Zero-variance intervals are descriptive, not proof of invariance. Programmatic references follow visible policies. Chinese policy/question/criteria text is authored; structural field and label IDs remain in English.

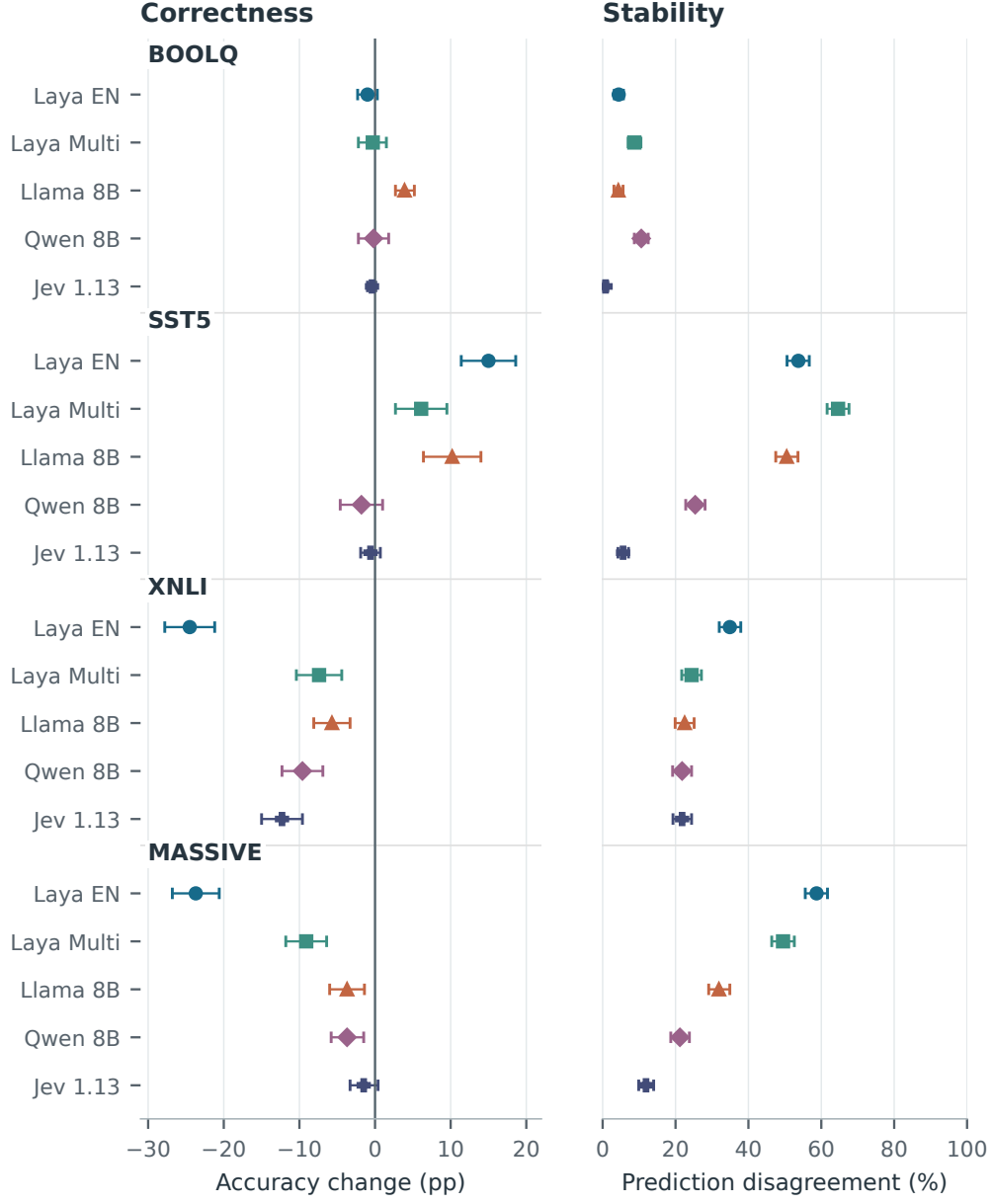


Figure 6: Exploratory paired interface contrasts against dataset-provided references on 1,000 aligned states per task: BoolQ choice minus noul, SST5 choice minus score, and Chinese minus English for XNLI and MASSIVE. Left: accuracy change; right: canonicalized prediction disagreement. Whiskers are pointwise 95% paired state-cluster bootstrap intervals (10,000 draws). Canonical label mappings are recorded explicitly; no marginal-CI subtraction is used.

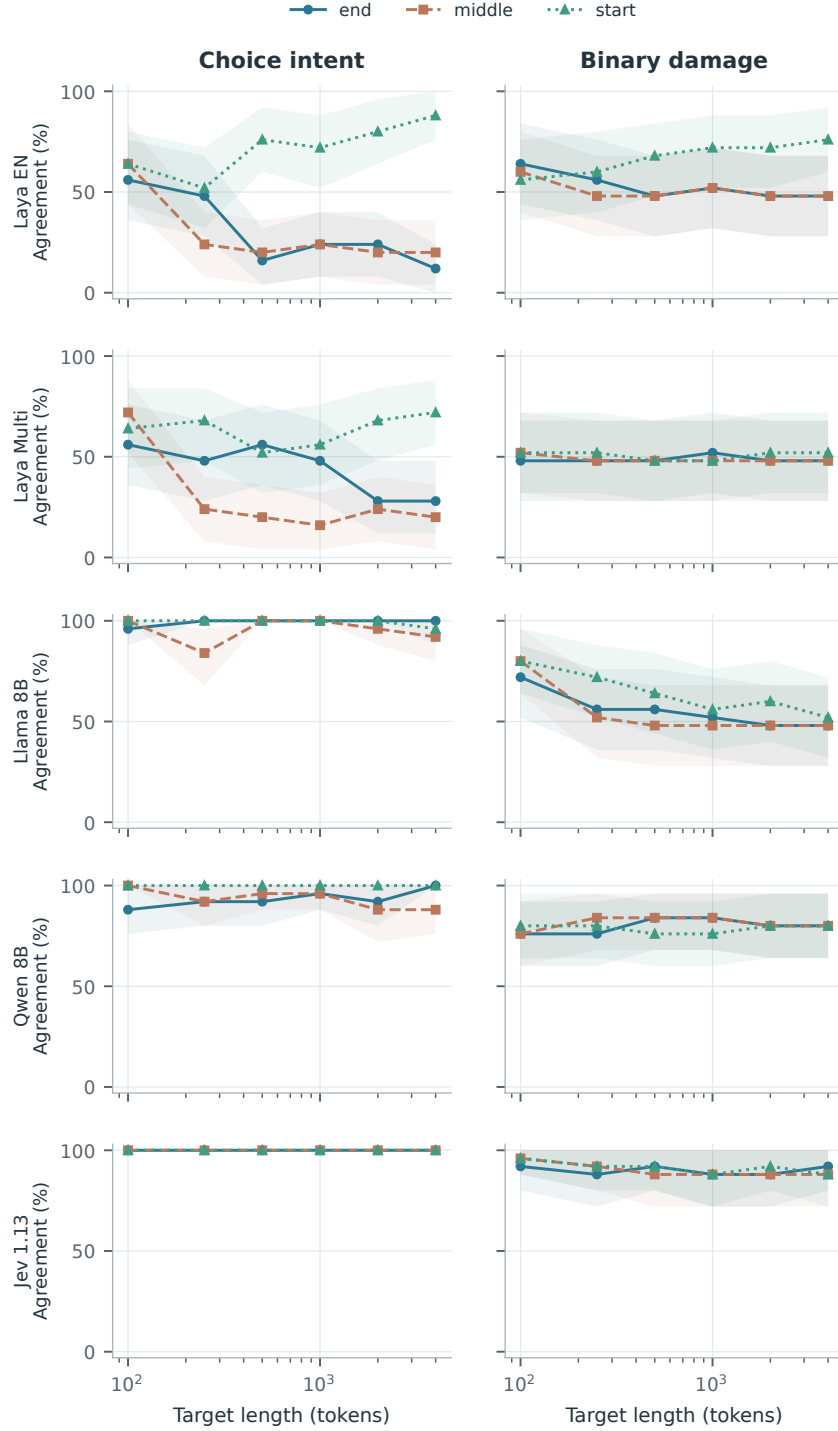


Figure 7: Programmatic needle reference agreement by source target length, insertion position and question. Rows identify models; left/right columns are the choice intent and binary damage questions. Each point uses 25 original needle clusters; shaded pointwise 95% cluster intervals resample those originals. Source length targets are not identical tokenizer token counts. The two primitives need not share the same context trend.

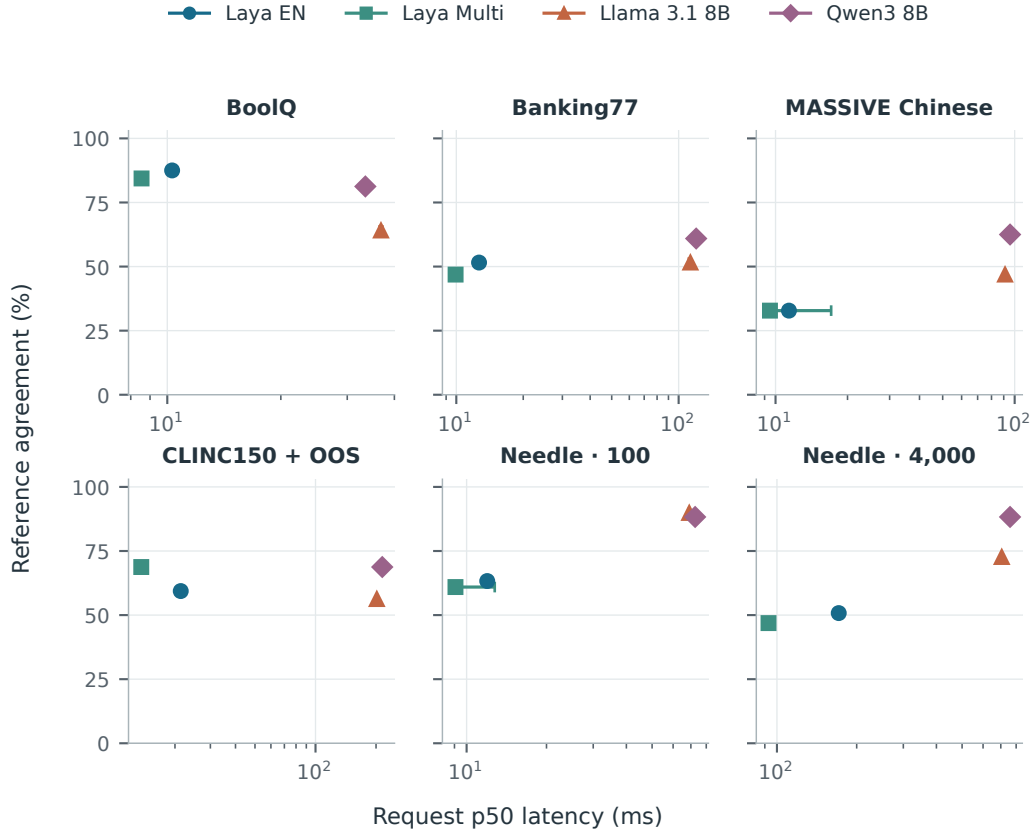


Figure 8: Resident-GPU quality/latency observations on six named workloads. Each stratum has the same 64 measured states across local models. BoolQ, Banking77, MASSIVE and CLINC150 use dataset references; the two needle workloads use programmatic references. Horizontal bars are 95% bootstrap intervals for the median of eight round p50 latencies; vertical coordinates are median reference agreement over repeated fixed states, not a new population-accuracy estimate. Jev’s hosted latency is excluded because its measurement boundary differs. No cross-workload frontier is drawn.

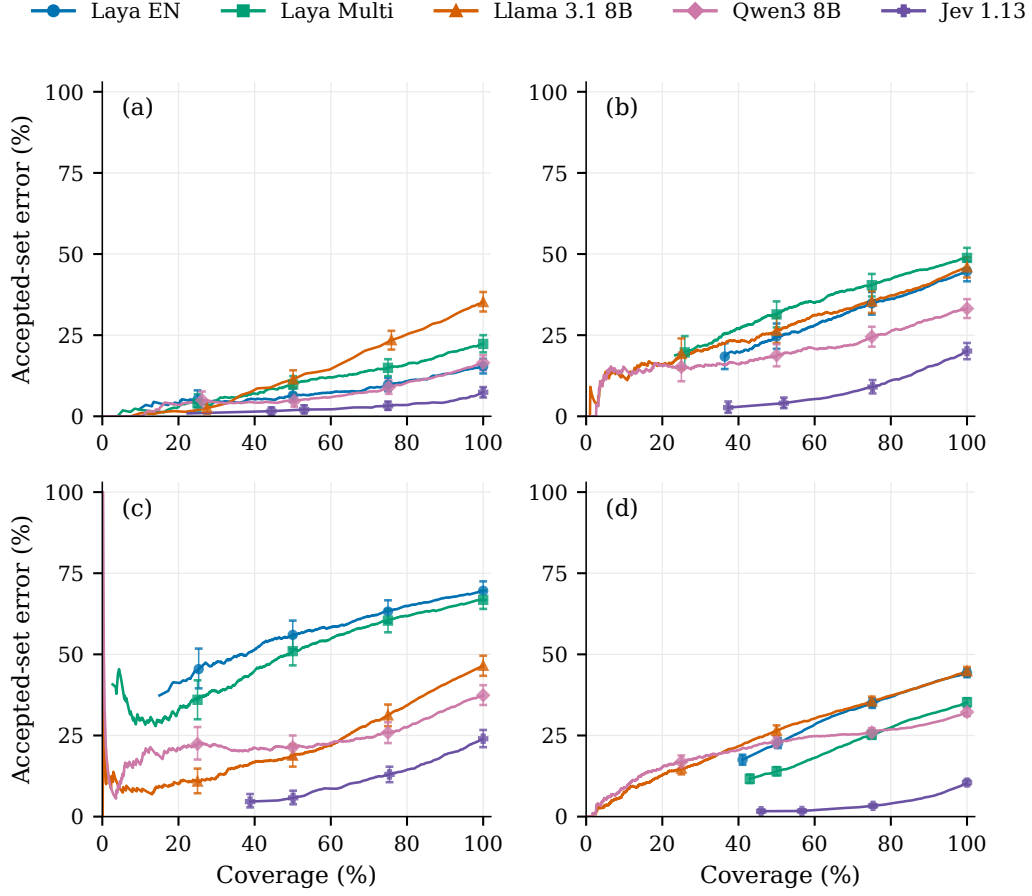


Figure 9: Maximum-probability selective error on (a) BoolQ, (b) Banking77, (c) MASSIVE Chinese and (d) CLINC150 with OOS (1,000, 1,000, 1,000 and 5,500 states). Ties are accepted together. Markers identify realized coverage at nominal 25/50/75/100% landmarks; intervals resample accepted state clusters conditional on the displayed threshold. Native and candidate-code probability semantics remain distinct. Curves describe the observed test set and do not validate a deployment threshold.

B Reproduction and data accounting

The historical accuracy record contains 22,934 logical requests and 26,450 decisions per model, including matched controls. The timing record contains 67,584 measured request executions and 98,304 decisions across repetitions. The fresh policy diagnostic contains 288 base states, 2,304 request variants and 6,912 decisions per evaluated model. The held-out wording test adds 96 new base states and 384 requests per model. These totals refer to different collections and replication units and should not be combined into an independent sample size.

The repository retains source/revision manifests, prepared-input fingerprints, model metadata, raw per-question probabilities and predictions, input audits, and complete timing/monitor records. The paired-analysis script checks raw-file fingerprints and reference alignment before deriving effects. Figure scripts consume its JSON output or the verified performance summary, producing vector PDF/SVG and PNG previews. The hosted runner reads credentials outside the repository, records every response attempt without authorization headers, pins the returned model ID, and resumes only under an unchanged contract.

Accuracy and typed-response validation can be recomputed from public outputs on CPU. New model inference additionally requires the pinned checkpoints or a provider account. Exact remote replication depends on the provider honoring its version identifier. Generator code and full programmatic confirmation cases are reproducible without downloading third-party state text.

Hosted execution deviation. The frozen retry description referred to 5xx errors generally, while the implementation enumerated HTTP 500, 502, 503, 504 and 529 (plus 429 and transport exceptions). A single HTTP 520 in the fresh refund Chinese condition received one attempt and caused three failed decisions, retained in primary denominators. Its Chinese-minus-original action effect is $-1/96$; a valid-pair sensitivity excludes that paired base state and yields $0/95$. This service failure does not support a language-regression claim. The historical matrix additionally contains four choice/probability-consistency failures affecting eight decisions, also retained. No failed answer was repaired or replayed.

One historical resume boundary also violates the protocol’s global manifest-order statement: two adjacent journal entries cross that ordering. The journal remains append-only, and keyed reconciliation verifies complete request identities, payloads and outputs. No scoring or selection depends on journal position; the original bytes and the execution deviation are preserved.